Sequential equilibrium (SE)

Let (σ, μ) be an assessment in a finite extensive form Γ.

Definition 1 (σ, μ) is a consistent assessment if there is a sequence of assessments $\{\sigma^n, \mu^n\}$ such that

1. $\forall i$, σ_i^n is a completely-mixed behavior strategy.
2. μ^n is obtained from σ^n using Bayes rule.
3. $\sigma^n \to \sigma$ in the Euclidean metric, and $\mu^n \to \mu$.

Definition 2 (σ, μ) is a sequential equilibrium if it is sequentially rational and consistent.

Example revisited

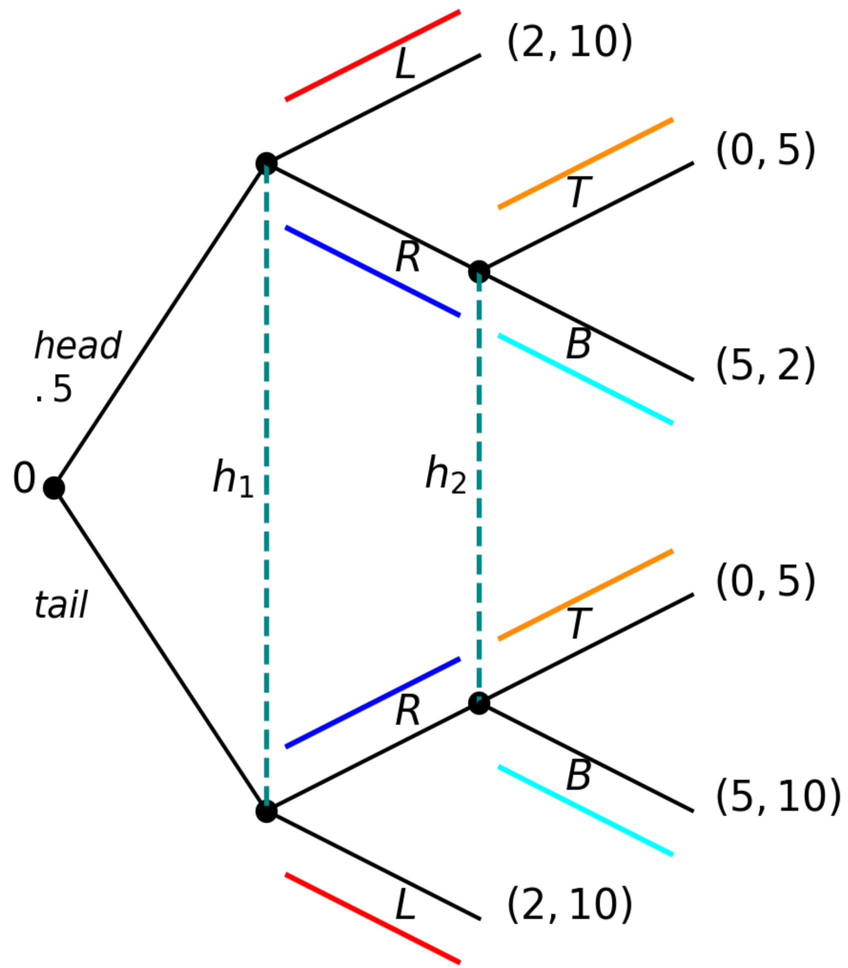

Figure 1

First WPBE and SE ($blue$). Affectionately (R, B).

$$\sigma_1(h_1) = (0,1);\ (L,R). \quad \mu_1(hd|h_1) = \frac{1}{2}$$

$$\sigma_2(h_2) = (0,1);\ (T,B). \quad \mu_2(hd,R|h_2) = \frac{1}{2}$$

Second WPBE but not SE (red). Affectionately (L, T).

$$\sigma_1(h_1) = (1,0);\ (L,R). \quad \mu_1(hd|h_1) = \frac{1}{2}$$

$$\sigma_2(h_2) = (1,0);\ (T,B). \quad \mu_2(hd,R|h_2) = \frac{9}{10}$$

(R, B) is a SE

- The proposed assessment is sequentially rational; it is a WPBE.
- It remains to show that it is consistent.

$$\sigma_1^n = \left(\frac{1}{n}, 1 - \frac{1}{n}\right),\ \sigma_1^n \to \sigma_1 = (0, 1)$$
$$\sigma_2^n = \left(\frac{1}{n}, 1 - \frac{1}{n}\right),\ \sigma_2^n \to \sigma_2 = (0, 1)$$
$$\mu_1^n(\cdot|h_1) = \left(\frac{1}{2}, \frac{1}{2}\right),\ d\mu_1^n \to \mu_1 = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$
\begin{aligned}
\mu_2^n((hd,R)|h_2) = \frac{P_{\sigma^n}(hd,R)}{P_{\sigma^n}(h_2)} \\
&= \frac{P_{\sigma^n}(hd)\sigma_1^n(R|h_1)}{P_{\sigma^n}(hd)\sigma_1^n(R|h_1) + P_{\sigma^n}(tl)\sigma_1^n(R|h_1)} \\
&= \frac{\frac{1}{2}\times(1-\frac{1}{n})}{\frac{1}{2}\times(1-\frac{1}{n})+\frac{1}{2}\times(1-\frac{1}{n})} \\
&= \frac{1}{2}
\end{aligned}
$$

Thus, $\mu_2^n \to \mu_2 = (\frac{1}{2}, \frac{1}{2})$.

(L, T) is not a SE

- Assessment (L, T) is sequentially rational; it is a WPBE.
- It remains to show that it is consistent. Let $x_n, y_n $ be any two sequences in $(0, 1)$ such that as $n \to \infty$, $x_n \to 0$ and $y_n \to 0$.

$$\sigma_1^n = (1 - x_n, x_n), \quad \sigma_1^n \to \sigma_1 = (1, 0)$$
$$\sigma_2^n = (1 - y_n, y_n), \quad \sigma_2^n \to \sigma_2 = (1, 0)$$
$$\mu_1^n(\cdot|h_1) = (\frac{1}{2}, \frac{1}{2}), \quad \mu_1^n \to \mu_1 = (\frac{1}{2}, \frac{1}{2})$$

$$
\begin{aligned}
\mu_2^n((hd, R)|h_2) &= \frac{P_{\sigma^n}(hd, R)}{P_{\sigma^n}(h_2)} \\
&= \frac{\frac{1}{2}x_n}{\frac{1}{2}x_n + \frac{1}{2}x_n} \\
&= \frac{1}{2}
\end{aligned}
$$

- Thus, $\mu_2^n \to \mu_2 = (\frac{1}{2}, \frac{1}{2})$.
- No perturbation leads to the desired beliefs.
- With $\mu_2 = (\frac{1}{2}, \frac{1}{2})$, player 2 cannot justify playing T.
- (L, T) is not a SE.

- The previous analysis shows that

$$\sigma_1(h_1) = (1,0);\ (L,R). \quad \mu_1(hd|h_1) = \frac{1}{2}$$
$$\sigma_2(h_2) = (1,0);\ (T,B). \quad \mu_2(hd,R|h_2) = \frac{9}{10}$$

  is not a SE

- Is there any SE in which (L,T) is played?

To answer, one must find beliefs μ_2 that justify player 2's moving T.

$$E[U_2|h_2,\sigma_1,T,\mu_2] \geq E[U_2|h_2,\sigma_1,B,\mu_2]$$
$$\mu_2(hd|h_2)5 + \mu_2(tl|h_2)5 \geq \mu_2(hd|h_2)2 + \mu_2(tl|h_2)10$$

$$
\begin{aligned}
E[U_2|h_2, \sigma_1, T, \mu_2] &\geq E[U_2|h_2, \sigma_1, B, \mu_2] \\
5 &\geq \mu_2(hd|h_2)2 + (1 - \mu_2(hd|h_2))10 \\
5 &\geq -\mu_2(hd|h_2)8 + 10 \\
\mu_2(hd|h_2)8 &\geq 5 \\
\mu_2(hd|h_2) &\geq \frac{5}{8}
\end{aligned}
$$

- Only beliefs $\mu_2(hd|h_2) = \frac{1}{2}$ are part of a consistent assessment with strategy profile (L, T).
- Thus (L, T) is never part of a SE.

Observations

Theorem 1 Let σ be an extensive-form (``trembling hand”) perfect equilibrium in an extensive-form game of perfect recall. Then σ can be completed to a sequential equilibrium: there is a belief system μ such that (σ, μ) is a sequential equilibrium.

Corollary 1 Every finite extensive-form game with perfect recall has a sequential equilibrium.

- See Maschler, Solan, and Zamir (2013), Theorem 7.57 and Corollary 7.58, page 280.

References

Maschler, Michael, Eilon Solan, and Shmuel Zamir. 2013. *Game Theory*. Translated by Ziv Hellman. 1st ed. Cambridge University Press.